Absolute error

1 State the absolute error of the following:

a A measurement of $30\ \text{kg}$, which was measured to the nearest kilogram.

b A measurement of $400\ \text{g}$, which was measured to the nearest gram.

c The amount $\$700$, which was calculated to the nearest $\$10$.

2 A chef measures the amount of olive oil to be used in a cup marked in millilitres. Find the absolute error of any measurement she makes.

3 State the absolute error of each of the following measurements obtained from measuring devices:

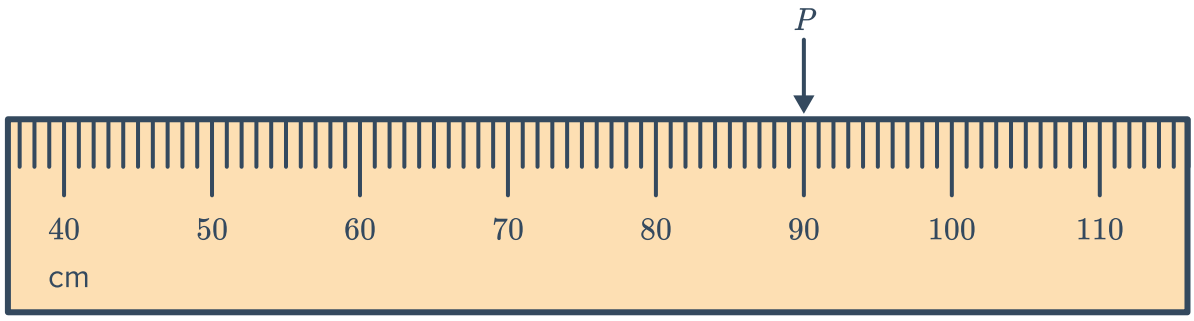
a $18.9\ \text{mL}$ **b** $269\ \text{mm}$ **c** $9.9\ \text{g}$ **d** $18\ \text{s}$

e $15.12\ \text{kg}$ **f** $17.8\ \text{m}$ **g** 117.38°C

4 The number of people at a concert is approximately 5050. Explain why we cannot find the absolute error of this approximation.

The limits of accuracy

5 Consider the tape measure below:



a What is the smallest unit marked on the tape?

b What should be recorded for measurement P?

c What is the absolute error of measurement P?

d State whether the following could be the actual measurement of P:

i $89.02\ \text{cm}$ **ii** $90.64\ \text{cm}$ **iii** $89\ \text{cm}$ **iv** $90.03\ \text{cm}$

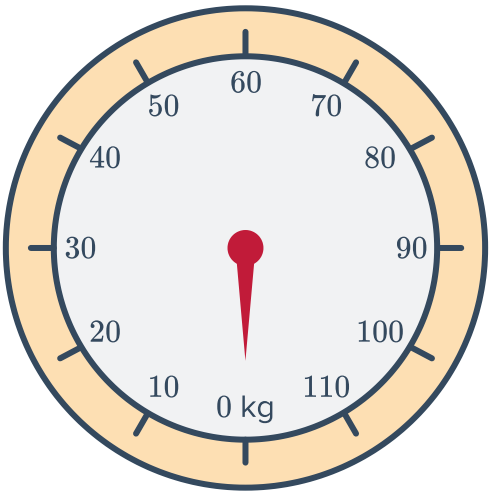
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

6 Consider the following scale:

a What is the smallest unit labelled on the scale?

b What is the absolute error?

c If a certain object is measured at $70\ \text{kg}$, what are the lower and upper bounds of this measurement?



7 Consider the speedometer below:

a At what speed is being indicated on the speedometer?

b What is the absolute error of this speed reading?

c What would the maximum speed limit need to be to ensure the driver is not speeding?



8 For the following measurements, find:

i The upper bound **ii** The lower bound

a A distance measured to be $13.45\ \text{km}$.

b A distance measured to be $6.4\ \text{km}$.

c A height measured to be $5\ \text{m}$.

d A bag of sugar weighs $14\ \text{kg}$ to the nearest 10 grams.

e Puncak Jaya, Indonesia's highest mountain, is $4884\ \text{m}$ high rounded to the nearest metre.

f The cost of a CD lie if it is known to be $\$50$ correct to the nearest $\$5$.

9 The length of a piece of rope is measured to be $19.99\ \text{m}$ using a ruler. What is the upper bound of the largest possible length of this rope?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 The height of a tower is measured as $2.1\ \text{m}$. What is the shortest possible height of the tower?

11 State the maximum and minimum possible number of the following:

- **a** A town's population is estimated to be 170 people, to the nearest 10 people.
- **b** A stack of paper is estimated to have 3000 sheets to the nearest 100 sheets.

12 $c = 4$ and $d = 8$ have both been rounded to the nearest whole number.

- **a** Calculate the lower bound of the value of $c \times d$.
- **b** Calculate the upper bound of the value of $\frac{c}{d}$.

13 A school field has a $100\ \text{m}$ track that is measured to the nearest metre. The record for the fastest $100\ \text{m}$ sprint is 13.9 seconds rounded to the nearest 0.1 seconds.

- **a** What is the fastest speed, in metres per second, the record holder could have run? Round your answer correct to two decimal places.
- **b** What is the slowest speed, in metres per second, the record holder could have run? Round your answer correct to two decimal places.

Percentage error

14 Calculate the percentage error of the following, correct to two decimal places:

- **a** A measurement of $55\ \text{kg}$, that is measured to the nearest kg.
- **b** A measurement of $5.2\ \text{km}$, that is measured to the nearest $0.1\ \text{km}$.
- **c** A measurement of $35\ \text{cm}$, that is measured to the nearest $5\ \text{cm}$.
- **d** A measurement of $410\ \text{mm}$, that is measured to the nearest $10\ \text{mm}$.
- **e** A measurement of $1700\ \text{mL}$, that is measured to the nearest $10\ \text{mL}$.

15 For the following scenarios, find:

- **i** The absolute error.
- **ii** The percentage error, correct to one decimal place.

- **a** A dishwasher weighs exactly $30.3\ \text{kg}$, but was measured by Luke to be $35\ \text{kg}$.
- **b** The length of a particular desk is exactly $1.33\ \text{m}$, but was measured by Tara to be $1.4\ \text{m}$.
- **c** The exact length of the lunch break taken by an employee was 52.2 minutes, but was measured by the employee to be 45 minutes.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

16 The number of civilian casualties in a particular war was exactly $83\,891$, but was assessed by Italy to be $68\,000$ and by England to be $100\,000$.

- **a** Find the absolute error of Italy's assessment.
- **b** Find the absolute error of England's assessment.
- **c** Find the percentage error of Italy's assessment, correct to one decimal place.
- **d** Find the percentage error of England's assessment, correct to one decimal place.
- **e** Which country's assessment was more accurate: Italy's or England's?

Area and volume calculations

17 A circle has a radius of $7\ \text{cm}$, correct to the nearest centimetre.

- **a** Calculate its area correct to one decimal place.
- **b** What are the smallest and largest possible areas? Round your answer to one decimal place.
- **c** What is the maximum possible absolute error in the calculation?
- **d** What is the maximum possible percentage error in the calculation? Round your answer to one decimal place.

18 A field has dimensions $15.4\ \text{m} \times 17.6\ \text{m}$, to the nearest $10\ \text{cm}$.

- **a** What are the upper and lower bounds of the area of the field?
- **b** What is the upper and lower bounds of the perimeter of the field?

19 A cube has a side length of $10\ \text{cm}$ rounded to the nearest centimetre.

- **a** Find the lower and upper bounds of the side length.
- **b** Find the area of one face of the cube, using the given side length.
- **c** Find the lower and upper bounds of the area of one face.
- **d** Find the volume of the cube using the given side length.
- **e** Find the lower and upper bounds of the volume of the cube.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647